

42.2 Orthogonal polynomials

The space $\mathcal{P}_d$, evaluated at the design, is an ordinary $(d+1)$ -dimensional subspace of $\mathbb{R}^n$. What is bad is the spanning set $1, x, \dots, x^d$. An orthonormal basis makes $\mathbf{X}^\top \mathbf{X} = \mathbf{I}$, the condition number one, and the coefficients independent of the degree — and, by Theorem 6.7.1, changes not one fitted value. Section 9.4 built these polynomials by Gram–Schmidt; this section supplies what they are for an arbitrary design, the recurrence that generates them in $O(nd)$ operations, and what changes when the basis is swapped.

42.2.1 The inner product of a design

Definition 42.2.1 — Orthogonal polynomials for a design

Let $x_1, \dots, x_n$ be design points, m of them distinct, with weights $w_1, \dots, w_n > 0$. For real functions f, g on the design write

$$\langle f, g \rangle = \sum_{i=1}^n w_i f(x_i) g(x_i), \quad \|f\|^2 = \langle f, f \rangle,$$

an inner product on the functions restricted to the design. Polynomials $p_0, \dots, p_d$ with $d < m$ are the **orthogonal polynomials of the design** if p_j has degree exactly j with leading coefficient one (it is **monic**) and $\langle p_j, p_k \rangle = 0$ for $j \neq k$. The normalized $q_j = p_j / \|p_j\|$ are the **orthonormal polynomials** of the design.

The restriction $d < m$ is the rank condition of Exercise (A1): with fewer distinct points some nonzero polynomial of degree d vanishes on the design, and the form is only nonnegative definite. Two facts make the construction work, and both are one line.

Lemma 42.2.1 — Multiplication by the argument is self-adjoint

For polynomials f, g let $(xf)(x) = xf(x)$. Then $\langle xf, g \rangle = \langle f, xg \rangle$.

Proof. Both sides equal $\sum_i w_i x_i f(x_i) g(x_i)$. ■

The second is that p_j is orthogonal to *every* polynomial of degree less than j , not only to $p_0, \dots, p_{j-1}$: those span $\mathcal{P}_{j-1}$, because their degrees are $0, 1, \dots, j-1$.

42.2.2 The three-term recurrence

Theorem 42.2.2 — Existence, uniqueness and the three-term recurrence

Let $d < m$. The orthogonal polynomials of the design exist, are unique, and are gener-

ated by

$$p_{-1} = 0, \quad p_0 = 1, \quad p_{j+1}(x) = (x - a_j)p_j(x) - b_j p_{j-1}(x), \quad j = 0, 1, \dots, d-1, \quad (42.2.1)$$

with

$$a_j = \frac{\langle xp_j, p_j \rangle}{\|p_j\|^2}, \quad b_j = \frac{\|p_j\|^2}{\|p_{j-1}\|^2} \quad (j \geq 1), \quad b_0 = 0.$$

Every $\|p_j\|^2$ with $j \leq d$ is strictly positive.

Proof. Existence and the recurrence together. Argue by induction on j , the claim being that $p_0, \dots, p_j$ produced by Eq. 42.2.1 are monic of degrees $0, \dots, j$, mutually orthogonal, and have positive squared norms. For $j = 0$, $p_0 = 1$ is monic of degree 0 and $\|p_0\|^2 = \sum_i w_i > 0$.

Suppose the claim holds up to $j \leq d-1$. The polynomial p_{j+1} defined by Eq. 42.2.1 is monic of degree $j+1$, since xp_j is monic of degree $j+1$ and the subtracted terms have degree at most j . Take inner products. First,

$$\begin{aligned} \langle p_{j+1}, p_j \rangle &= \langle xp_j, p_j \rangle - a_j \|p_j\|^2 - b_j \langle p_{j-1}, p_j \rangle \\ &= \langle xp_j, p_j \rangle - a_j \|p_j\|^2 = 0 \end{aligned}$$

by the choice of a_j and the induction hypothesis. Next, for $j \geq 1$, using Lemma 42.2.1 and $\langle p_j, p_{j-1} \rangle = 0$,

$$\begin{aligned} \langle p_{j+1}, p_{j-1} \rangle &= \langle p_j, xp_{j-1} \rangle - a_j \langle p_j, p_{j-1} \rangle - b_j \|p_{j-1}\|^2 \\ &= \langle p_j, xp_{j-1} \rangle - b_j \|p_{j-1}\|^2. \end{aligned}$$

Now xp_{j-1} is monic of degree j , so $xp_{j-1} - p_j$ has degree at most $j-1$ and is therefore orthogonal to p_j . Hence $\langle p_j, xp_{j-1} \rangle = \|p_j\|^2$, and the choice of b_j makes the display vanish. Finally, for $k \leq j-2$, Lemma 42.2.1 gives

$$\begin{aligned} \langle p_{j+1}, p_k \rangle &= \langle p_j, xp_k \rangle - a_j \langle p_j, p_k \rangle - b_j \langle p_{j-1}, p_k \rangle \\ &= \langle p_j, xp_k \rangle = 0, \end{aligned}$$

because xp_k has degree at most $j-1 < j$. So p_{j+1} is orthogonal to all of $p_0, \dots, p_j$.

It remains to show $\|p_{j+1}\|^2 > 0$ when $j+1 \leq d$. If it were zero then p_{j+1} , a monic polynomial of degree $j+1 \leq d < m$, would vanish at all m distinct design points, which is impossible.

Uniqueness. Suppose $\tilde{p}_0, \dots, \tilde{p}_d$ also satisfy the definition. Since the degrees are $0, \dots, j$, the set $\{p_0, \dots, p_j\}$ is a basis of $\mathcal{P}_j$, so $\tilde{p}_j = \sum_{k \leq j} c_k p_k$. Orthogonality of $\tilde{p}_j$ to $\mathcal{P}_{j-1}$ gives $0 = \langle \tilde{p}_j, p_k \rangle = c_k \|p_k\|^2$ for $k < j$, hence $c_k = 0$; and both being monic forces $c_j = 1$. ■

The recurrence needs only the two previous polynomials, so the work is $O(nd)$ rather than Gram-Schmidt's $O(nd^2)$; it never forms x^j for $j > 1$, so nothing overflows or cancels; and a_j, b_j describe the basis in $2d$ numbers, which is how a fitted polynomial should be stored.

Remark (Why only three terms). Writing $xp_j = \sum_{k \leq j+1} c_{jk} p_k$, the coefficient $c_{jk} \|p_k\|^2 = \langle xp_j, p_k \rangle = \langle p_j, xp_k \rangle$ vanishes whenever xp_k has degree below j , that is whenever $k < j-1$. The same argument gives three-term recurrences for the classical families — Legendre, Chebyshev, Hermite, Laguerre — the orthogonal polynomials of $\langle f, g \rangle = \int fg w$, which [Section 39.5](#) used for numerical integration.

42.2.3 Sequential fitting

Corollary 42.2.3 — Fitting in the orthonormal basis

Let $\mathbf{q}_j$ have entries $q_j(x_i)$ and let the weights be one, so the $\mathbf{q}_j$ are orthonormal in $\mathbb{R}^n$; write $\hat{\gamma}_j = \mathbf{q}_j^\top \mathbf{y}$. For every $d < m$:

- (a) the least squares fit of degree d is $\hat{f}_d(t) = \sum_{j=0}^d \hat{\gamma}_j q_j(t)$, and $\hat{\gamma}_j$ does not depend on d ;
- (b) $E(\hat{\gamma}_j) = \langle f, q_j \rangle$ when $E(Y_i) = f(x_i)$ for any function f , and $\text{Cov}(\hat{\gamma}) = \sigma^2 \mathbf{I}$;
- (c) the residual sum of squares equals $\|\mathbf{y}\|^2 - \sum_{j \leq d} \hat{\gamma}_j^2$, so the sequential sum of squares for degree j is $\hat{\gamma}_j^2$ on one degree of freedom, free of the order of entry;
- (d) the F statistic for $H : \gamma_d = 0$ in the model of degree d is $\hat{\gamma}_d^2 / \hat{\sigma}_d^2$, with $\hat{\sigma}_d^2$ that model's residual mean square, on $(1, n - d - 1)$ degrees of freedom.

Proof. With orthonormal columns $\mathbf{X}^\top \mathbf{X} = \mathbf{I}$, so $\hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{y}$ componentwise, which is (a); the columns for degrees below d do not change when d grows, as in Corollary 9.4.2. Part (b) is $E(\mathbf{q}_j^\top \mathbf{Y}) = \mathbf{q}_j^\top \boldsymbol{\mu} = \langle f, q_j \rangle$ with $\text{Cov}(\mathbf{X}^\top \mathbf{Y}) = \sigma^2 \mathbf{X}^\top \mathbf{X}$; part (c) is Pythagoras, $\|\mathbf{y}\|^2 = \sum_{j \leq d} \hat{\gamma}_j^2 + \|\hat{\boldsymbol{\varepsilon}}\|^2$, and Theorem 9.4.1; part (d) is Theorem 11.6.1 for a single coefficient with $\text{Var}(\hat{\gamma}_d) = \sigma^2$, squared. ■

Part (b) is worth a second look. Whatever the true regression function, $\hat{\gamma}_j$ estimates the j th coefficient of its expansion in the design's own orthonormal basis, with variance σ^2 for every j . Where the sequence sinks into the noise is where the data stop distinguishing degrees: the practice of choosing the degree from the sequential sums of squares, and its honest statement.

Warning. Stopping at the first small $\hat{\gamma}_j^2$ is selection from the same data: the chosen model's residual mean square is too small, its intervals too narrow, and the F statistic that justified the stop is not F (Proposition 29.5.1). The sequence can also dip and rise again, as it does below, so one insignificant term is a weak reason to stop.

42.2.4 The Nile, revisited

Example. (The recurrence on the Nile series)

The annual flow of the Nile at Aswan, 1871–1970, gives $n = 100$ equally spaced design points, rescaled here to $[-1, 1]$. Example (A trend in the Nile) built its orthonormal

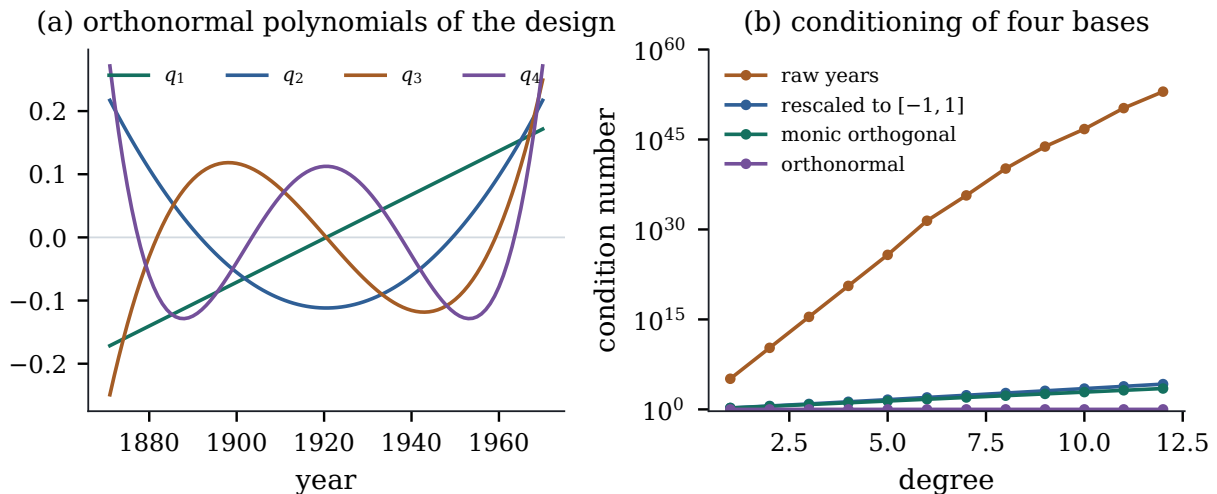


Figure 42.2.1: (a) The orthonormal polynomials $q_1, \dots, q_4$ of the Nile design, from the three-term recurrence. (b) Condition number of four bases for that space: raw years, rescaled years, monic orthogonal, orthonormal.

polynomials by a QR factorization of the matrix of powers; Eq. 42.2.1 gives the same basis, the largest discrepancy between corresponding columns being 1.1×10^{-14} and the largest inner product between distinct columns 6.7×10^{-16} , machine precision. The design being symmetric about its mean, every a_j vanishes (Exercise (B1)), and the first four values of b_j are 0.34007, 0.27197, 0.26213, 0.25871; Figure 42.2.1(a) draws the polynomials themselves.

The sequential sums of squares reproduce the table of Example (A trend in the Nile) — 613893, 309415, 1894, 134142 for degrees 1 to 4, and 4225 and 7034 for degrees 5 and 6 — with F statistics 27.08, 15.70, 0.10, 7.18, 0.22 and 0.37. The cubic term is negligible and the quartic is not: a reader stopping at the first insignificant term would have missed it.

```
import numpy as np
import statsmodels.api as sm

nile = sm.datasets.nile.load_pandas().data
year = nile["year"].to_numpy().astype(float)
y = nile["volume"].to_numpy()
n = len(y)
u = 2 * (year - year.min()) / (year.max() - year.min()) - 1    # design rescaled to [-1, 1]

def orthogonal_polynomials(t, d):
    """Monic polynomials  $p_0, \dots, p_d$  orthogonal over the design  $t$ , by the
    recurrence."""
    P = np.zeros((len(t), d + 1))
    P[:, 0] = 1.0
    a, b, sq = np.zeros(d), np.zeros(d), [float(len(t))]
    for j in range(d):
        a[j] = (t * P[:, j] @ P[:, j]) / sq[j]
        P[:, j + 1] = (t - a[j]) * P[:, j]
        if j > 0:
            b[j] = sq[j] / sq[j - 1]
            P[:, j + 1] -= b[j] * P[:, j - 1]
```

```

        sq.append(P[:, j + 1] @ P[:, j + 1])
    return P, a, b, np.array(sq)

P, a, b, sq = orthogonal_polynomials(u, 8)
Q = P / np.sqrt(sq)                                # orthonormal
        columns
print("a_j:", np.round(a[:5], 6))
print("b_j:", np.round(b[1:5], 5))
print("largest inner product between distinct columns:",
      f"{np.max(np.abs(Q.T @ Q - np.eye(9))):.2e}")

```

[Run this code online](#)

```

coef = Q.T @ y                                     # gamma_j = <q_j, y>, unchanged by
        the degree
seq_ss = coef[1:] ** 2                             # sequential sum of squares for
        each degree
sse = np.sum((y - y.mean()) ** 2) - np.cumsum(seq_ss)
for j in range(1, 7):
    F = seq_ss[j - 1] / (sse[j - 1] / (n - j - 1))
    print(f"degree {j}: SS = {seq_ss[j - 1]:9.0f}    SSE = {sse[j - 1]:9.0f}    F = {F
      :7.2f}")

```

[Run this code online](#)

Remark (A limit that explains the numbers). For a design filling an interval evenly, $n^{-1} \sum_i f(x_i)g(x_i)$ is a Riemann sum for $\frac{1}{2} \int_{-1}^1 fg$, so the design's orthogonal polynomials approach the monic Legendre polynomials, with $a_j = 0$ and $b_j = j^2/(4j^2 - 1)$ ([Exercise \(B2\)](#)): that is 0.33333, 0.26667, 0.25714, 0.25397 against the Nile's 0.34007, 0.27197, 0.26213, 0.25871. Since $b_j \rightarrow 1/4$, the monic polynomials shrink geometrically: badly scaled even though orthogonal, so normalizing matters as much as orthogonalizing.

42.2.5 What changes and what does not

For the Nile design the condition numbers of four bases for $\mathcal{P}_d$ are

Degree	raw years	rescaled	monic orthogonal	orthonormal
2	1.83×10^{10}	3.70×10^0	3.29×10^0	1
4	3.90×10^{20}	1.85×10^1	1.26×10^1	1
6	2.83×10^{31}	9.77×10^1	4.92×10^1	1
8	1.42×10^{40}	5.31×10^2	1.93×10^2	1

Orthogonality by itself buys a factor growing from about one to about three over this range; the orthonormal basis has $\mathbf{X}^\top \mathbf{X} = \mathbf{I}$, so its condition number is one at every degree and least squares in it is a matrix–vector product. The gap between the last two columns is entirely scaling.

What does *not* change is everything else: $\hat{\mathbf{Y}}$, $\hat{\boldsymbol{\varepsilon}}$, every leverage, every band, every F statistic about the *space*. The degree-16 polynomial of [Figure 42.1.1](#) oscillates identically in the orthonormal basis, and its extrapolation is still nonsense. Orthogonal polynomials answer the first complaint of [Section 42.1](#) and are silent on the second.

Remark (Where these have appeared before). For a factor with quantitative levels, the same construction with weights $w_k = n_k$ gives the polynomial contrasts of [Section 8.7](#) (Example (Trend in ideology across party identification)): the orthogonal polynomial coding of a factor and the basis that software offers for a continuous regressor are one object.

42.2.6 Exercises

42.2.6.1 A. Check your understanding

Exercise A1.

Compute the monic orthogonal polynomials p_0, p_1, p_2 for the design $x = (-1, 0, 1)$ with unit weights from [Eq. 42.2.1](#), and check that their integer-scaled versions are the contrasts $(-1, 0, 1)$ and $(1, -2, 1)$.

Exercise A2.

A quartic is fitted in the raw powers and again in the orthonormal polynomials of the design. Which of these differ between the two fits: the linear coefficient, the fitted value at x_1 , the residual sum of squares, the sequential sum of squares for degree 3, the leverage h_{11} , the condition number of $\mathbf{X}$?

42.2.6.2 B. Practice

Exercise B1.

Show that if $x \mapsto 2c - x$ permutes the design and preserves the weights, then $a_j = c$ for every j and $p_j(2c - x) = (-1)^j p_j(x)$.

Solution. Take $c = 0$ after a shift, and induct. p_0 is even, and $p_1 = x - a_0$ with $a_0 = \langle x, 1 \rangle / \|1\|^2 = 0$ by symmetry, so p_1 is odd. If p_{j-1} and p_j have parities $(-1)^{j-1}$ and $(-1)^j$, then $x p_j^2$ is odd, so $a_j = 0$, and $p_{j+1} = x p_j - b_j p_{j-1}$ is a sum of two polynomials of parity $(-1)^{j+1}$.

Exercise B2.

The monic Legendre polynomials are orthogonal for $\langle f, g \rangle = \int_{-1}^1 f g$. Using [Theorem 42.2.2](#) and the identity $\int_{-1}^1 P_j^2 = 2/(2j+1)$ for the standard Legendre polynomials, whose leading coefficient is $\binom{2j}{j} 2^{-j}$, verify that $b_j = j^2/(4j^2 - 1)$.

Solution. Write $\tilde{P}_j = P_j/\ell_j$ with $\ell_j = \binom{2j}{j} 2^{-j}$ for the monic version. Then $\|\tilde{P}_j\|^2 = 2/\{(2j+1)\ell_j^2\}$ and

$$b_j = \frac{\|\tilde{P}_j\|^2}{\|\tilde{P}_{j-1}\|^2} = \frac{(2j-1)\ell_{j-1}^2}{(2j+1)\ell_j^2}.$$

Since $\ell_j/\ell_{j-1} = (2j-1)/j$, the ratio is $(2j-1)j^2/\{(2j+1)(2j-1)^2\} = j^2/(4j^2-1)$.

Exercise B3.

Given $a_0, \dots, a_{d-1}$, $b_1, \dots, b_{d-1}$, $\|p_j\|^2$ and $\hat{\gamma}_0, \dots, \hat{\gamma}_d$, write down an algorithm that evaluates $\hat{f}_d(t)$ at a new t in $O(d)$ operations and storage, without ever computing a power of t . Why is this preferable to expanding $\hat{f}_d$ in the monomial basis and using Horner's rule?

Solution. Run [Eq. 42.2.1](#) forwards at the single point t , accumulating $\sum_j \hat{\gamma}_j p_j(t)/\|p_j\|$ as you go; only $p_{j-1}(t)$ and $p_j(t)$ need be kept. (Running it backwards, accumulating the coefficients instead, is Clenshaw's algorithm, the standard way of summing a Chebyshev or other orthogonal-polynomial series.) Horner's rule is equally fast but needs the monomial coefficients, and converting the fit to those is the ill-conditioned step: the conversion matrix is the change of basis whose condition number the table above reports.

42.2.6.3 C. Going deeper

Exercise C1.

Show that p_j has j distinct real zeros, all strictly inside the range of the design. (Suppose it changes sign at only $r < j$ interior points $z_1, \dots, z_r$, and consider $\langle p_j, \prod_l (x - z_l) \rangle$.)

Exercise C2.

For two regressors the analogue of $\mathcal{P}_d$ is the span of the monomials $x^r z^s$ with $r + s \leq d$. List an orthogonal basis $\phi_1, \phi_2, \dots$ of it in a fixed order of nondecreasing total degree. The one-variable proof works because $x p_j$ expands in p_{j-1}, p_j, p_{j+1} alone; exhibit a design in $\mathbb{R}^2$ and an index i for which $x \phi_i$ has a nonzero inner product with some ϕ_k , $k < i - 1$, so that no scalar three-term recurrence generates the list. (Vector recurrences over a whole block of one total degree do exist; the single-variable bookkeeping is what fails.) Relate this to [Chapter 44](#).